

WESTERN SYDNEY UNIVERSITY



SCHOOL of: Engineering, Design and Built Environment

ASSIGNMENT COVER SHEET

STUDENT DETAILS

Name: Eylia Wahbeh

Student ID: 20374600

SUBJECT AND TUTORIAL DETAILS

Subject Name: Interdisciplinary Global Home

Subject code: ENGR4021

Tutorial Group: Shasha stars

Day: 7/5/2025

Time: 11:00

Lecturer or Tutor name: Sasha Alexander

ASSIGNMENT DETAILS

Title: Functional Evaluation

Length: 1000 words + group

Due Date: 7/5/2025

Date submitted: 7/5/2025

Home campus (where you are enrolled): Parramatta city.

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Project Brief:

Our space design scenario will develop and innovate upon existing information and concepts of what can improve astronauts' health and habitat whilst on Long-distance missions aboard the International Space Station (ISS). The team aims to improve the astronauts' interaction aboard the ISS whilst on missions, particularly the longer expeditions into outer space. As a team, we've focused our investigations on three main aspects of their interaction, which include: habitat design, psychological & physical wellbeing and nutrition practices. Ultimately, we aim to create an

environment that optimises and enhances the performance of the universal space missions whilst benefiting the needs of astronauts.

Refined Team Project Brief (G):

Health, Health & Habitat: Influential Project

Domain 1: Space on Long-Distance Missions

1 - Ergonomics

The ergonomic interactions and space aboard the ISS have been identified as a major concern for astronauts exposed to a microgravity environment on LDM's. Extended exposure to weightlessness has been shown to lead to bone density loss, circulation issues and muscle atrophy. The fitness wearable I aim to create will monitor vital, muscle/bone density and circulation continuously 24/7 to the medical professional in charge of the astronauts' health on Earth. They will collect this data and create solutions for these concerns and communicate this information through Claire's device to ensure the astronauts maintain good health. The living quarters aim to reduce the musculoskeletal strain astronauts experience during missions. These implementations aim to control and mitigate the ergonomic risks associated with LDM's on the ISS identified in our investigation from A1.

2 - Exercise

Exercise is a critical consideration in space as it is the major challenge astronauts need to overcome to maintain their physical health whilst aboard the ISS. On Earth, gravity aids the human body's function, however, when exposed to space microgravity, it causes the deterioration of bones and muscles. The major concern is that the human heart works less effectively. When the body is not exercising, these issues can cause heart strain, bone density loss and muscle atrophy whilst on LDM. While on missions, Astronauts are required to complete 2.5 hours of exercise daily to maintain and reduce the listed health risks (Australian Space Agency, n.d.). This exercise is done using machines like treadmills, weight lifting and exercise bikes, while tethered to the ISS to maintain physical health.

3 - Technology

Ergonomics of the ISS is strictly linked to technology as it considers how these systems aid the workflow aboard the space station and the impact they place on the astronauts. Technology aboard the ISS provides 'social interactions for astronauts' to communicate between the space station and their loved ones on Earth, creating a 'network' (Grouport, n.d.). It will be utilised and improved in **Claire's** idea to enhance the existing social communication links aboard the ISS. Additionally, tech will be used in my fitness wearable (product name) to provide live feedback to astronauts and health professionals on Earth to improve the maintenance of astronauts' health.

4 - Accessibility

Accessibility aboard the ISS needs to be improved to increase the workflow and safety on LDM's. Our team has identified these flaws and has implemented these design considerations into our part of our space scenario to enhance comfort, safety, work ethic and

efficiency. When implemented, listing everyone's products will improve mobility and usability aboard the ISS and support astronauts in completing daily tasks.

5 - Mental Health

The impact of LDM's on astronauts' mental health is significant. The main cause is loneliness. A study was conducted by Lunark, which identified the psychological effects when confinement leads to a strong desire for social interaction. This resulted in disruptive behaviours amongst crew and symptoms of depression and homesickness (Riva et al., 2022). This has been detrimental in the past and resulted in poor overall efficiency on space missions due to the toxic atmosphere and conflict/frustration amongst the crew. (Claire's product and mine) Prioritise strategies to improve the environment on the ISS to promote a calming and welcoming atmosphere to minimise loneliness and enhance the psychological health of astronauts. This will lead to increased focus and productivity when completing work on LDM's.

6 - Modularity

Modularity is an important consideration when discussing LDMs as it allows for the customisation of a design per use case scenario, which is particularly relevant when dealing with the limited confines of space within the ISS. As such, modularity should be considered where possible, especially when designing spaces to be lived in. In our redesigns, Christian aims to utilise modularity as a feature to enhance experiences in shared living spaces (allowing astronauts to interact with their environment in ways which encourage crew interaction), whilst Ben's use of modularity focuses on enhancing personal spaces (allowing astronauts to customise their own sleeping quarters).

7 - Sleep (Christian and Ben)

To remain productive and sane in a working position and general life, the quantity and quality of sleep must be maintained to maintain an active position without taking risks of damage, mentally and physically. The main factors include a regular cycle of sleep, such as maintaining or simulating a day-night cycle, and having a stable amount of sleep per rest around 8+ hours.

8 - Privacy (Christian and Ben)

Privacy in close quarters locations such as the ISS is important as it provides opportunities to destress and time for mental rest. While studies suggest that people are generally "lonelier and less satisfied on days in which they spent more hours in solitude" (Weinstein, et. al, 2023), these lonely feelings have been shown to be nullified and even reduced when solitude was an individual's decision (Weinstein, et. al, 2023). As such, Ben's redesign allows astronauts the agency to choose exactly how they enjoy their privacy — an astronaut should have access to locations of privacy, and these spaces to be accessible and of high quality. Without these, the astronauts can feel claustrophobic and can suffer from mental breaks from the pressures of work and colleagues

9 - Socialisation (Christian and Ben)

In addition to privacy, socialisation is another core consideration when discussing astronaut wellbeing on LDMs as it is an inherent aspect of the human experience. Research states that “the functional capacity and structure of the brains and nervous systems can be strengthened by cultivating caring relationships supported by self-care strategies” (Jordan, 2023). This demonstrates the importance of socialisation within the ISS, as it benefits astronauts’ mental capacity, enabling them to complete tasks in a better frame of mind. Moreover, the development of positive, high-quality social relationships has been known to be beneficial to the mental health of individuals (Andersen et. al, 2021). As such, Christian’s concepts focus on developing interesting environments which promote collaboration and interactiveness.

10 - Reminders of Earth (Christian and Ben)

Reminders of Earth are important to consider as a feature of astronaut life in space. As space life is a foreign experience that may feel isolated or lonely, these connections to Earth are imperative to maintain healthy living in the ISS.

Examples of current Earth reminders within the ISS include (see images below):

- Hatch doors decorated with stickers and photographs from past missions
- Prepackaged meal choices/options
- Cultural traditions — holidays are celebrated (eg, Christmas)



(Nespoli and Miller, n.d)

8 Standard Menu Categories packed in BOBs



(Douglas, n.d)



(NASA, as cited in Rosenbluth, 2024)

Such reminders allow astronauts to maintain a sense of familiarity and normalcy in such an unfamiliar living and working environment. These reminders are core elements of Ben and Christian's designs, as both feature concepts that allow astronauts to customise their environment to feel more homely.

11 - Maintenance and machinery longevity (Eyliya and Dang)

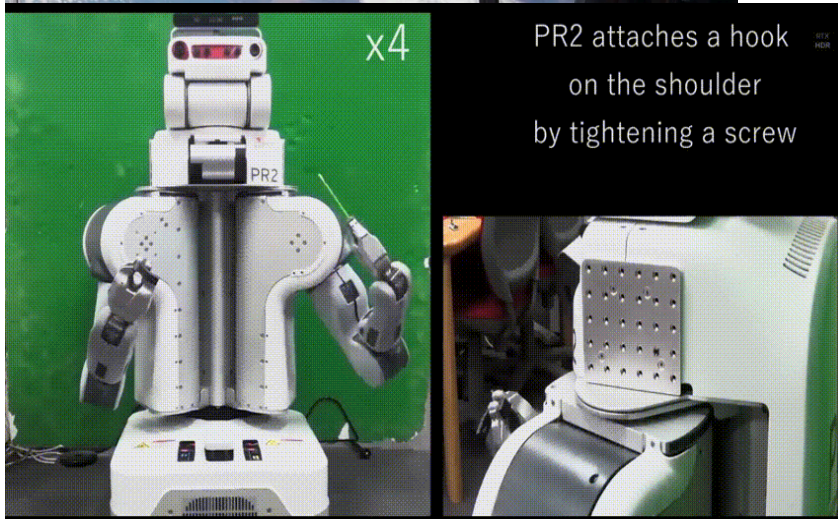
Automation requires maintenance by astronauts on board. Lack of mobility caused by lack of gravity can easily increase risk and safety during repair and maintenance inside any self-moving facility. Microgravity also makes repair and maintenance more challenging in space due to how materials behave differently.

Without gravity, fluid can leak during maintenance when attempting to pour or contain it. Therefore, lubrication and adhesives will also be a hassle to work with.



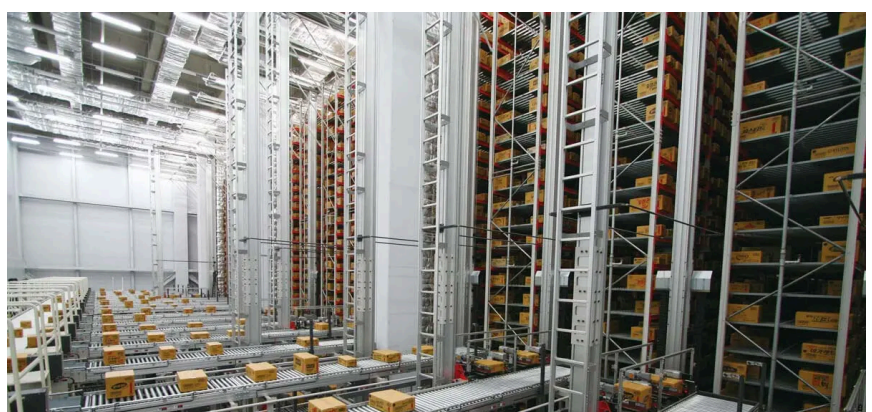
The materials are now weightless, which is a good thing because it is now easier to move heavy parts around by hand.

But for smaller equipment, they can also wander around and lodge themselves into machinery, increasing the chance of a catastrophic failure.



Leveraging sensor and self-diagnostics software, and a potential robot design that can perform self-repair (Pescovitz, 2019). This can provide some level of maintenance, but always be prepared for any need of human intervention.

12 - Storage management (Eyliya and Dang)



Storing food can either be integrated into each chamber or create a dedicated room for all food supply, seedlings.

The bigger the storage size, the more maintenance and time are required to constantly organise and relocate each different resource on a daily basis. The lack of manpower will make it impossible without automation.

The storage design is an essential factor helping with packing and covering all the hanging and floating items on the wall, releasing the stress of a messy environment.

13 - Safety (Eylia and Dang)

This is a major factor to consider due to being in a small, compact space and to having zero gravity. In the ISS having food crumbs is a hazard for astronauts' safety as it will float into their eyes. My proposed design "VR" uses a backend safety net to control sound, lighting, weather and scent, which ensures it complies with NASA's safety guidelines, eliminating any long term risk.

We need to have consideration for the surrounding facility with easy access and an observatory for the astronauts when accessing the automated facility. Each must have a kill switch for the worst-case scenario. Software will cease all functions once it detects any member entering the operating zone.

14 - Microgravity (Eylia and Dang)

With nothing holding everything down, we can either generate artificial gravity or use the lack of gravity to our advantage. Vegetation and meat will be affected by microgravity during the production phase.

A storage system requires an encapsulating design, ensuring all supplies are tightly packed and held down at all times

Many reports support the astronauts loving the feeling of freedom that microgravity provides. But with that, it also provides hidden challenges affecting their mental health by preventing them from practising their daily earthly routine.

15 - Food Quality (Eylia and Dang)

Food quality will be constantly monitored and inspected through each process cycle, from the first phase of production, cultivation, to raw materials. Raw materials will be kept separately and each category to prevent cross-contamination. Cooking can be done via machine or standard cooking from the astronauts, completely up to their liking.

Evidence-based support (G):

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Review Team Tasks (G):

Week 5

Claire: We began this week reviewing the previous assessment. We also took the class time as an opportunity to show our individual design developments from the first assessment. Key takeaway was having a better understanding of each other's scope of the project.

Adrian: We discussed and reviewed our previous assessments. We also used this class time to showcase our individual concept developments from A1.

Ben: We had a discussion and presentation of our previous assignment, and designated class time to objectively analyse them for future assignments, and created a project scope

Dang: We have limited vision on the whole scope of the next project, but we started by revising the previous assessment.

Eylia: I was off during the class time, but I connected with our team chat, getting all updates: Assessment Two Intro and How do the different projects from A1 go together

Christian: This week was mainly a week of reflection on our submission, and a week to ideate for potential ways to move forward.

Week 6

Claire: This week, we focused on refining the project brief. We also had a collaborative conversation about delegating the group components of the assessment as well as narrowing the most influential attributes for this project.

Adrian: This week we refined our group project brief. We also had collaborative conversations about the hypothetical group space for our project.

Ben: Refinement of the assignment and group tasks were given out for attributes, and the beginning of proposal sketches and ideas

Dang: We narrow down our objective, reducing inaccuracy on our individual task and begin to properly brainstorm our design

Eylia: I was off this week for being sick
Team updated, they did a bit of ideation and sketches for the group task

Christian: We designated our key considerations to be divided up, so that the group work was split evenly. Our group selected the most important considerations from our previous assignment to make a consolidated list.

Week 7

Claire: As the University break is approaching, we decided to discuss the plan and expectations for what each of us will be working on to ensure progress is maintained for the assessment

Adrian: We discussed what needs to be completed from now to week 9 in preparation for our submission.

Ben: We discussed what would happen over the period between now and the due date, aiming to balance the individual and group portions of the assignment to reduce load and stress

Dang: We mentioned each other one more time before break, made sure everyone is aware of the deadline.

Eylia: We discussed the layout of the space station and planned how our projects could go together

Christian: This week, we arranged to keep in contact over the following week's break.

Week 8

Claire: We had a break

Adrian: Completed 3 concept CAD files.

Ben: No work was done

Dang: I created the first prototype design for an automated agriculture using centrifugal force as artificial gravity, with footage simulating movement and functionality in Solidworks. Not much work was done

Eylia: planned out my design project and did deep research, outlining all key features

Christian: Week off

Week 9

Claire: Following week 8 break, we dedicated an in-depth and significant amount of time discussing how we will integrate our individual project ideas together. We explored several forms and shapes, aiming to find a cohesive design that meets our brief parameters

Adrian: After returning, we had a deep discussion about the time we would need to combine our concepts. We explore many iterations for potential ideas and confirm brief parameters for our areas.

Ben: We discussed our current standings and understanding of our project in depth about integration and inclusion of each other's ideas into a single vessel, with much discourse and integration methods, and discussion for future assignment 3

Dang: We reviewed each member's progress, made a few corrections, set boundaries for the overall scale, and made sure our current design is integrable with each other's design for the next assessment 3.

I also created the second and third prototype designs for an artificial biome and a bioreactor

Eylia: progress review for each member, outline key features to link our individual progress to the team's contribution and to the main project

We spoke a bit about assessment 3 and how we will get there

Christian: We had an in-class sketching and brainstorming session. Finalised certain ideas, reconsidered others, and generally just refined upon our ideas and concepts.

Week 10 - Deadline

Claire: During submission week, we all discuss the finalised presentation slides and what we would like to demonstrate to each other and our class.

Adrian: Submission week and presentation.

Ben: Submission and presentation slides

Dang: Submit to the lord and present ourselves in class.

Eylia: Submission and presentation slides

Christian: Submission time

Further Discipline Support review (G)

Identify the people you would want to work with

Adrian:

An Ergonomics/Human factors Engineer would have significantly contributed to creating Adrian design as they specialise in designing within the capabilities and limitations of the human body. This knowledge would be crucial when creating jewellery that would be worn for long periods, thus needing to be comfortable for the daily movement of the users.

Another discipline that would be involved in providing great insights would be an Aerospace Medicine Physician, as they specialise in the medical components of flight and space travel, which would give good indicators on what Adrian device design should track and collect data for as a health device. To optimise the success and improve the astronaut experience.

Claire:

A discipline that need to be included to ensure that the mood tracker design is accurately meeting the needs of the astronauts and can positively be implemented into the crews interactions with their habitation cycle is a psychologist as they focus on the human's mind and behaviour which would highlight the data that would needed and how to interpret it to be successful.

Subsequently, technical experts like software and hardware engineers would need to be used and discussed to aid in creating Claire's design, as they have more knowledge on the digital application of tracking the data in the most effective form. A device that is supposed to aid in stabilising and monitoring the emotions of the crew needs to work seamlessly to eliminate frustration or resistance to use.

Ben:

Sleep Scientists - A sleep scientist would allow a greater knowledge and understanding of the patterns of sleep an astronaut would have and the effects of 0 gravity within a space designed for it, allowing for more depth of discussion and necessary features that could be missed out

Astronautical engineers - Allows for greater extension of abilities and constructs in space that could broaden the ideas that can be generated and apply better practices that would be used on current rockets

Dang:

A **structural engineer** will help design the facility for the outer space environment, ensuring vibrations from the facility don't affect other structures. The design they make can take advantage of or protect against microgravity and shield us from radiation. Any part of the facility that fails to maintain proper seal and pressure can lead to a **total catastrophic breakdown**, compromising the entire space station.

Along with a **mechanical engineer** to design any moving components, including the rotating centrifuge mechanism, and automated machinery. This team will be exposed to the **highest fatality rate** tasks, including physical intervention and maintenance, both from inside and outside of the space station.

Control by the software required a **software developer** to create a custom interface and algorithm to synchronise all operations. They will be responsible for making any bug fixes during maintenance. Faulty software can lead to machinery breakdown and unpredictable movement, which can lead to the **death** of the astronauts when physical intervention is being performed.

During the process of food making, a **biotechnologist** can take advantage to study plant biology and their behaviour, monitor and study the microbes associated with the plant system. Mistakes from faulty sensors can cause contamination to enter the food supply, potentially **poisoning** the crew on board. The team will also help analyse and manage the chemical aspects of the growing environment, allowing for proper plant and meat growth, ensuring the crops and meat grown will be nutritious and safe for astronauts to eat.

Eylia:

Psychologists - The whole project is based on psychological factors; therefore, Psychologists are needed to study different VR scenarios and their effect on human behaviour. The study will not only be placed on the general influence of a scenario it should also provide a very detailed understanding of every individual reaction to an event. This will later allow us to boost astronauts' mood with an automated system that understands individuals' psychological

Human Factors and Crew Health Specialists - Define environmental standards, ensuring that NASA-STD-3001 are followed with regard to lighting, noise, and air quality. With the help of a Health Specialist, we will be able to create a new framework, verifying that VR/dining modules meet health and safety requirements, outlining and eliminating any potential hazards

Wearable Technology and Data Analytics Experts - Develop wearable sensors and interfaces that will be used to activate and change ship control

Technicians / VR specialists - Configure and install the designed system, mand ap the configuration to ensure seamless communication

Christian:

Engineers: A wide range of engineers would be required to ensure the success of this redesign project:

- *Mechanical:* to ensure that certain designs are feasible and realistic to create
- *Electrical:* to effectively and efficiently route electricity throughout the designs
- *Software:* to develop/update software and systems so they fit the context of spaceflight

User Experience Designers: Involving UX designers in the development moving forward would greatly benefit this project, ensuring that the experiences of Astronauts are considered alongside the technical and mechanical considerations.

Astronauts: In addition to UX designers, astronauts themselves would be crucial to be involved in the development process, as these are going to be the users of the 'end-product'. By incorporating the users into the development process, we ensure that our redesigns effectively address the needs of our users.

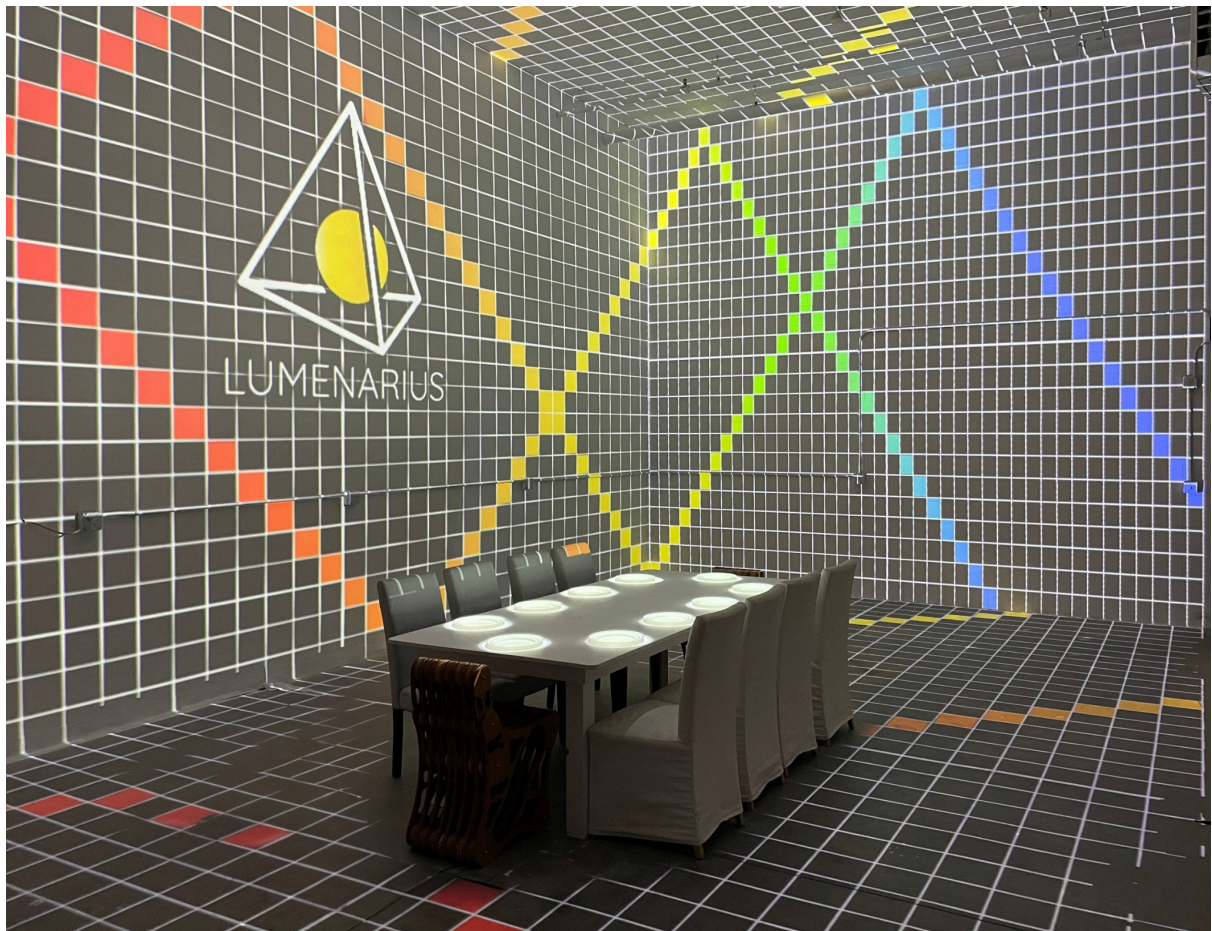
Psychologists: When dealing with designing spaces for wellbeing and social interactions, it is valuable to understand how designs can influence mental health. As such, engaging with psychologists may provide important insight into the development of shared environments for improved health and well-being.

TimeLine:

Click the link below (screenshot is also available in our presentation submission):

https://docs.google.com/spreadsheets/d/1j8y9v4I2mkGBp1jRzU6VKKbSm9_i8DND0iArC10XRyQ/edit?gid=1115838130#gid=1115838130

VR ROOM



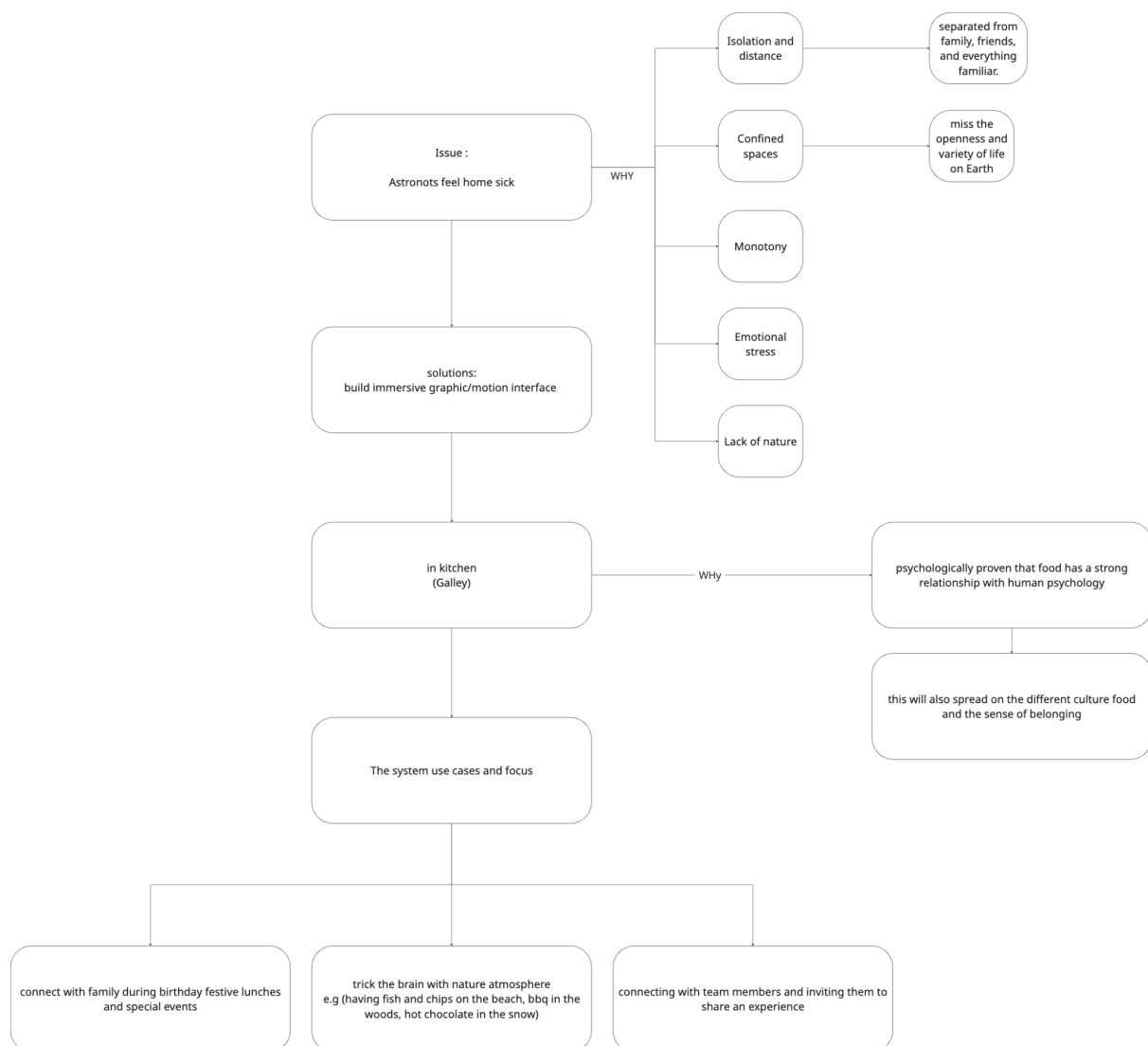
Concept Development

System proposal

Objective:

Tricking the astronaut mind to think and feel the warmth of earth

To achieve this a system plan was put in place with the use of an immersive graphic/motion interface. Below is an outline of the cause and solution



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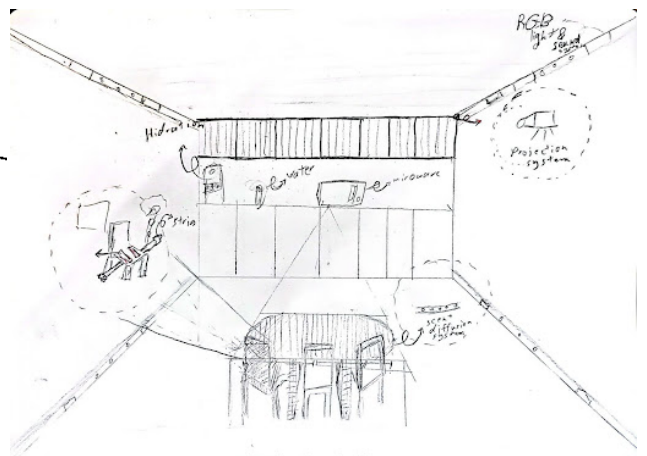
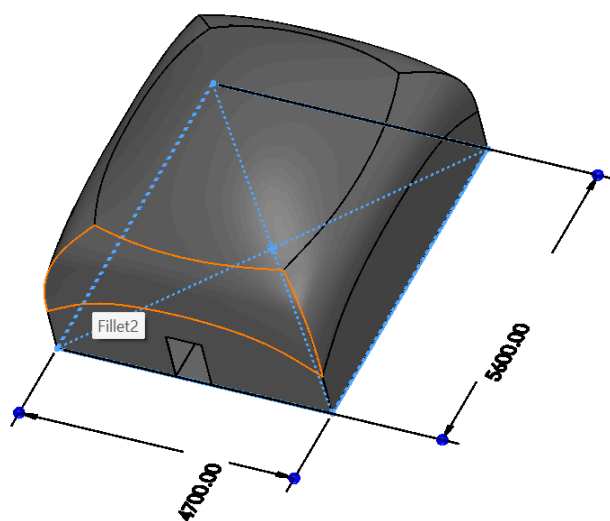
Method

The system is set to be planed out in the ISS kitchen with the potential of expanding across the whole ship

The conceptual design will be mainly focused on the astronauts mental status during food time, with the success of the concept this can expand across the whole ISS.

The kitchen of the ISS is called Galley

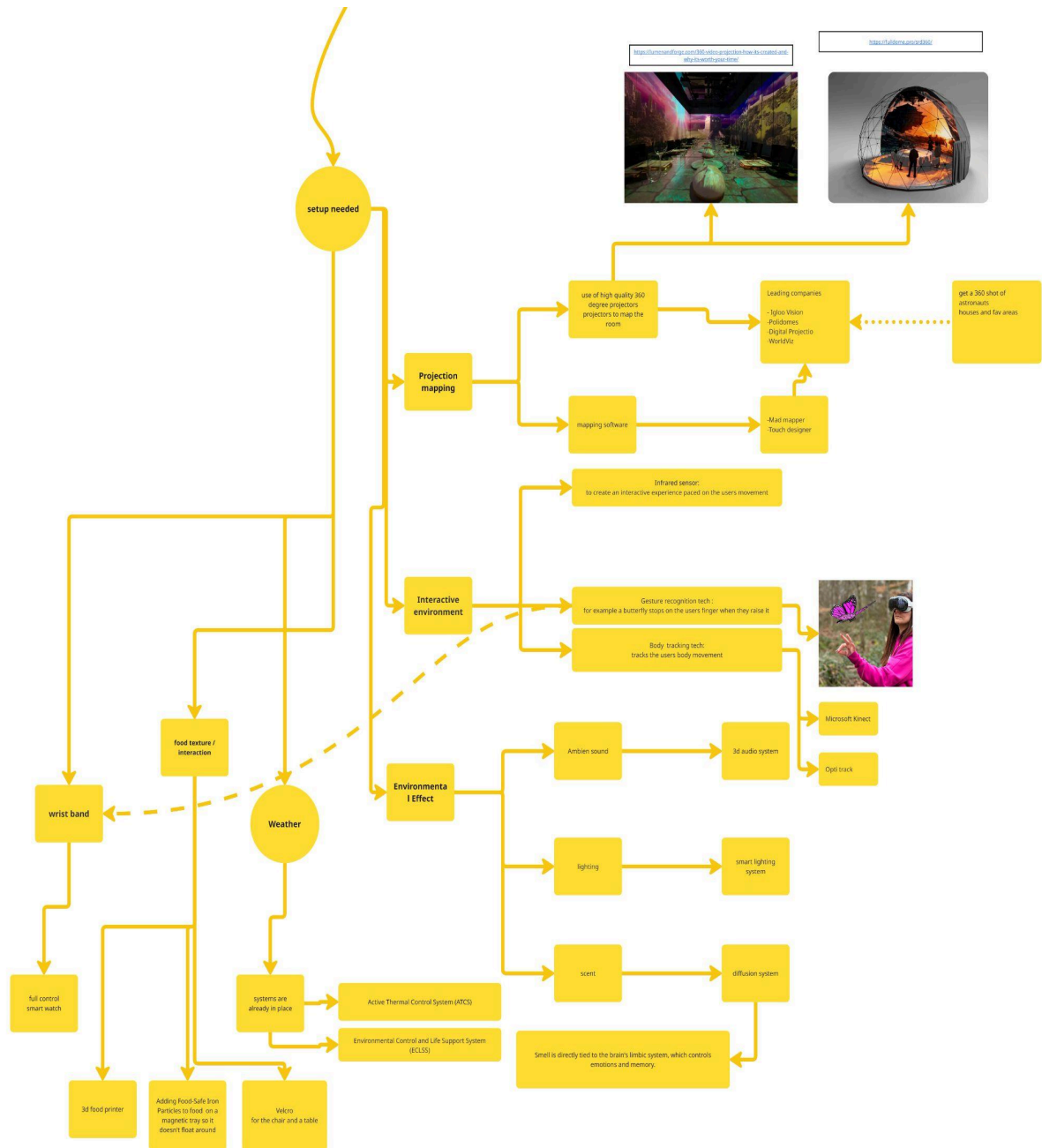
The galley is located in the clean areas of the ship and will have a an area of $5.6 \times 4.7 = 26.32 \text{ M}^2$



Astronauts will be able to use all different senses to simulating a real life experience

The room will be setup with

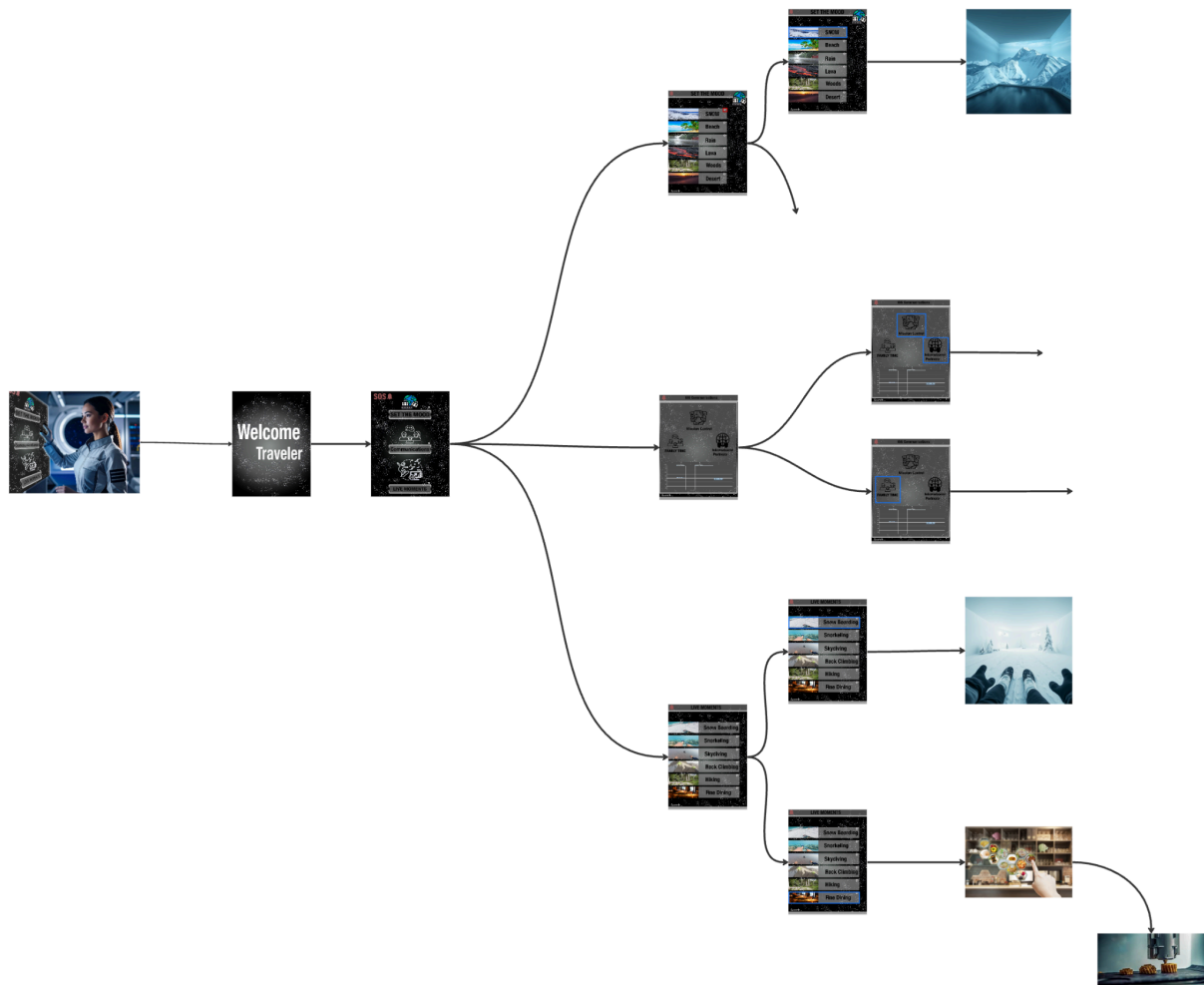
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3 different concepts

Concept 1

Screen projected hologram interface

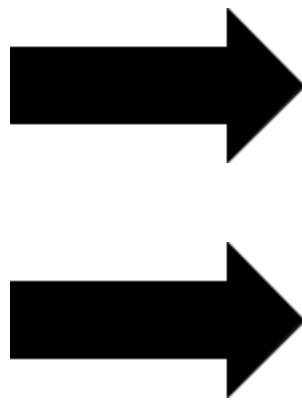


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Concept 2

Full voice command controlled

Flashing blue light when the keyword (Hey Jarvis) is used to activate voice control



Command Keycodes

Function	Natural Language Example
Start VR	"Launch VR sim 3"
Change temperature	"Set temp to 22°C in deck 1"
Adjust lighting	"Dim lights in medbay"
Play ambient sound	"Play ocean waves in quarters"
Ventilation control	"Increase airflow in lab"
Safety check	"Run safety protocol on deck 3"
Diagnostics	"Scan system status"
Reminder	"Remind me to hydrate at 1600"
Sleep mode	"Initiate sleep mode"
Help	"What can you do?"

Conversation Scenario:

Jackson:

Jarvis, launch the Mars terrain sim in VR.

JARVIS:

Launching Mars Terrain Simulation in Virtual Reality. Syncing vitals and environment. Would you like environmental conditions to match Mars average?

Jackson:

Yes, match the simulation.

JARVIS:

Temperature adjusted to -60°C, red-filter lighting enabled, sound set to Martian wind. Sim begins in 5... 4... 3...

Jackson:

Jarvis, stop wind sound, and make it a bit warmer.

JARVIS:

Disabling wind audio. Raising temperature to -30°C for comfort. Confirm?

Jackson:

Confirm.

Concept 3

Wearable activated system

This combines both of the previous concepts into one wearable responsible in analysing the data entered by a human and translating it into action through an NFC touch point and gesture control

This concept details are pending on the completion of wearables design by “Adrian”



New Space Habitat & Astronaut Experience (I)

The design in the kitchen will be used 3 to 4 hours per astronaut and can overlap

Breakfast (~45 mins)

Lunch (~45 mins)

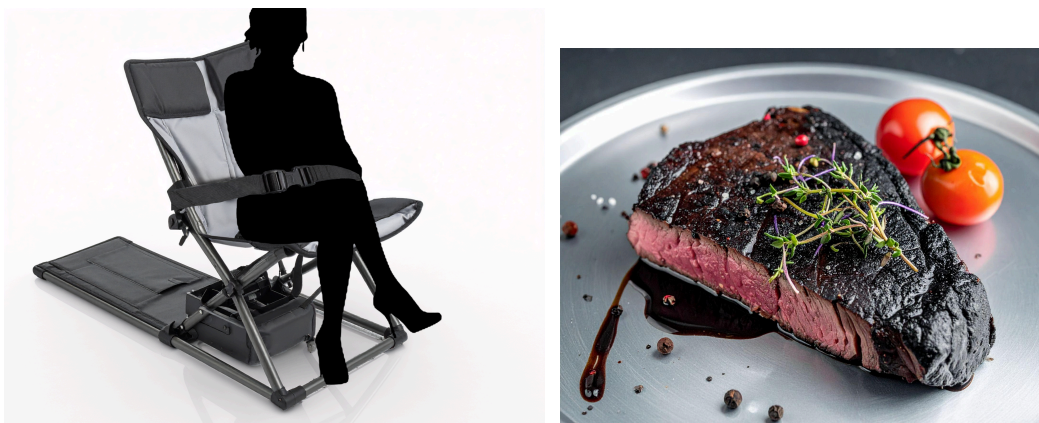
Dinner (~1–1.5 hours)

Extra time for snacks or baking

This can increase with the use of VR as this will be a place where people can share moments with others.

Design for zero gravity is considered

The proposed design will include tightened chairs & magnetic table and consumable metallic food particle



The provided space will alter the habitat to match the astronaut Emotional and Psychological state.

1- The simulation can recreate personal space from earth offering comfort and familiarity
As multiple reports indicate astronauts feeling emotional distance and disconnection.

2- VR or pre-recorded call of family gathering and special events reducing the feeling of isolation. As Cady Coleman reported in her interview “ there is no reason to come home if not for family “ (Youtube.com, 2025) and Butch Wilmore missing last year of his daughter's high school (Chung, 2025).

3- relaxing scenarios VR environments like beaches, forests, or guided meditation spaces can provide much-needed psychological relief in the confined and artificial setting of the ISS, Scott Kelly “being on a space station is difficult they have to be always ready not much of comfort, misses the fresh air , and can't imagine living isolated for so long without nature (trip to mars)” (Astronaut Scott Kelly,2016)

Visual and tactile cues (I)

The system is designed to have a minimal to no visual and tactile visibility. With the world of interior design moving toward headen smart feachers , lighting , sound ... , the proposed design will only come visible when interacting with. The visible cues of the system are flashing blue lighting and a projected display , and sound response system.

Designing Human Centred Systems

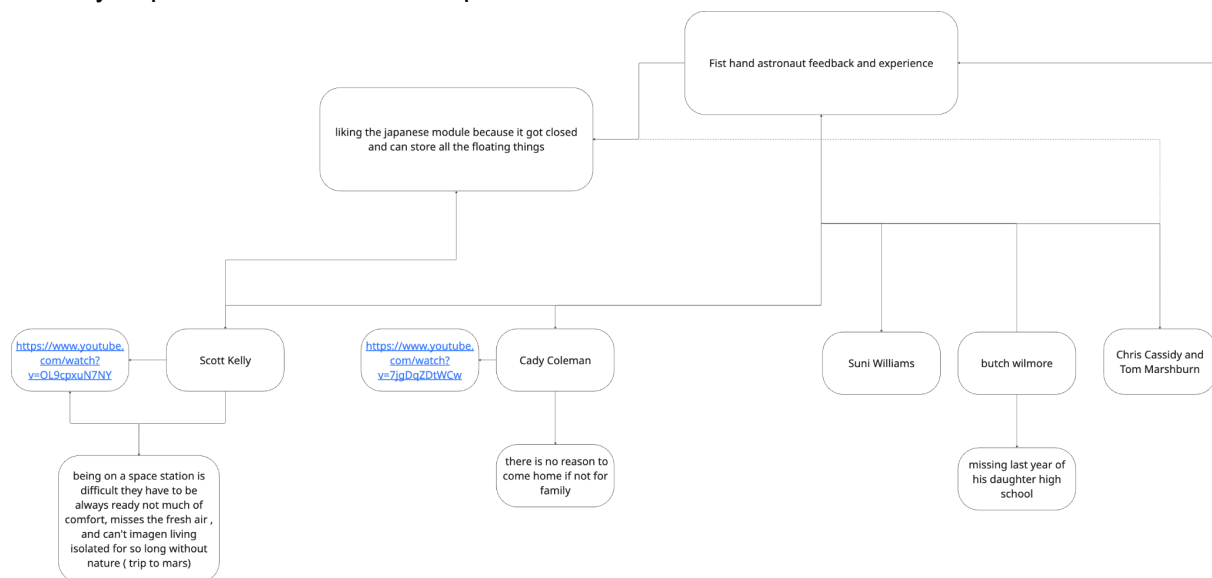
Crew Interaction Area

The interface will let astronauts book rooms and time for personal or shared use. This helps improve communication and emotional connection between team members. Using VR, they can share experiences or stories with each other. For example, on Christmas, they could gather in VR to connect with the station and their families, exchange gifts, or enjoy virtual settings especially useful during communication blackouts.

The system will have the ability to 3d print food for special occasions.

Human Factors verification

Based on human-centered design and supported by research in space psychology (Kanas & Manzey, 2008), the VR room provides immersive simulations of Earth-like environments—such as green fields, oceans, or time with loved ones—to reduce stress and sensory deprivation. This is further proven with astronaut feedback as shown below .



Through the system safety net not allowing extreme conditions that could endanger the astronauts health in any way will not be allowed.

	Requirements/Stand ards	Why Appropriate as Safety Net for VR Rooms	References
Sound Loudness (Noise)	- Maintain noise levels $\leq$ NC-50 (general areas) - Quiet zones $\leq$ NC-40 (e.g., private spaces)	Controlled noise reduces distractions, prevents hearing damage, and lowers stress, supporting user focus and comfort during VR immersion.	NASA-STD-3001 Volume 2 (Acoustic Limits) https://ntrs.nasa.gov/api/citations/20200002973/downloads/20200002973.pdf Health & Safety Guide https://www.classvr.com/policies/health-and-safety/
Lighting	- Daytime lighting: cool-white (~6500 K) - Pre-sleep or low	Proper, adjustable lighting reduces eye strain and motion	NASA-STD-3001 Volume 2 (Lighting) https://ntrs.nasa.gov/api/citations/20200002973/downloads/20200002973.pdf

	activity: warm-white (~2700 K) - Task lighting ≥ 500 lux	sickness, improves comfort, and enhances immersion in VR environments.	Indoor Environmental Quality Guidelines https://sftool.gov/learn/about/1/indoor-environmental-quality-ieq
Environmental Conditions ("Weather")	- Temperature: 18.3–26.7°C (65–80°F) - Relative humidity: 25–70% - Maintain safe CO2 levels	Stable temperature and humidity prevent discomfort and health risks during VR sessions, especially in enclosed rooms with multiple users.	NASA-STD-3001 Volume 2 (Environmental Health) https://ntrs.nasa.gov/api/citations/20200002973/downloads/20200002973.pdf VR Usage Guidance https://jtc1info.org/augmented-and-virtual-reality-safety-guidance-on-safe-immersion-set-up-and-usage/
Odor Control	- Continuous air filtration and purification to minimize odors and contaminants	Odor control maintains a pleasant environment, reducing distraction and discomfort for VR users in shared or prolonged sessions.	NASA-STD-3001 Volume 2 (Air Quality) https://ntrs.nasa.gov/api/citations/20200002973/downloads/20200002973.pdf Indoor Air Quality Standards https://sftool.gov/learn/about/1/indoor-environmental-quality-ieq

Holistic: My Contribution to LDM Approach (I)

My suggestion fits in nicely with other people's work. Its overall value is increased by incorporating design elements from various team members. For instance, Dang's suggestion to grow food in space enhances my design by providing astronauts with fresh ingredients, which would greatly enhance their dining experience. Similar to this, Adrian's fitness tracker can be used to both monitor the astronauts' health and act as a control system for the virtual reality experience.

The proposed design falls well within the scope of all key areas. For mental well-being, it helps astronauts stay connected to Earth, offering relaxation and a sense of connection to family. For physical health, the VR system can be integrated into a gym setting to motivate and enhance exercise routines. And food will ensure the best dining experience.

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